School Of Computing and Informatics

Website Design and Development

Assignment 1

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1. DNS

1.1 Types And Purpose Of DNS

Before discussing the types and purposes of DNS we must discuss what DNS is first. DNS or Domain Name System is a networking system that do mapping between the Ip addresses and the domain name, the domain name is a unique identifier for the Ip address, if we type the domain name in the URL the DNS converts it to an Ip address for the computer to understand. The DNS is a large number of servers, organized in a hierarchical fashion and distributed around the worlds, it have a root-level domains, country level domains, top-level domains, secondary level domains, and service, for example, www.htu.edu.jo, www is the service, htu is a secondary level domain, edu is the top-level domain, and jo is the country level domain.

DNS have various types such as:

- Recursive resolvers or recursor resolver: a server that acts like the middleman between user and other DNS servers, it is ta server that receives a queries from the user and the recursive resolver will lookup to find the Ip address that is connected to the domain name, if the Ip address is saved, the server give the Ip address to the user, if the Ip address is not saved, the server send a request to other DNS name servers to find the Ip address.
- Root name server: It is a server that acts like the library that points to the next step, it is the server that the Recursive resolver server sends a request to if the recursive resolver server does not have the Ip address. The root name server doesn't know the Ip address but it sends the user query to the right Top-Level Domain name server.
- TLD (Top-Level Domain) name server: it is a server that manages request based on the domain extension such as (.com, .org, .edu, etc.), and when the query arrive to the Top-Level Domain server and finds out the extension, the query goes to the Authoritative name server to lookup for the specific domain.
- Authoritative name server: it is the final step in the DNS lookup process, it provides the Ip address of the requested domain name and saves the DNS record for the specific domain name.

An important aspect of the DNS is the DNS lookup, the DNS lookup translates the domain name into an Ip address that the computer understands.

After discussing the types of DNS we must discuss the purposes of DNS, the DNS does mapping between Ip addresses and domain names, it converts domain names to IP addresses. Also, because it is hard for humans to know all

Ip addresses of websites, it is easier for us to know the domain name, so the DNS makes it easier for humans to navigate on the internet.

Another purpose of the DNS is that makes the internet user-friendly, reliability, and accessibility, it allows users to access websites from anywhere in the world.

1.2How Domain Names Are Organised and Managed

Domain Names are managed in a hierarchical structure, we have the root level, country-level, top-level, secondary-level, and service or hostname.

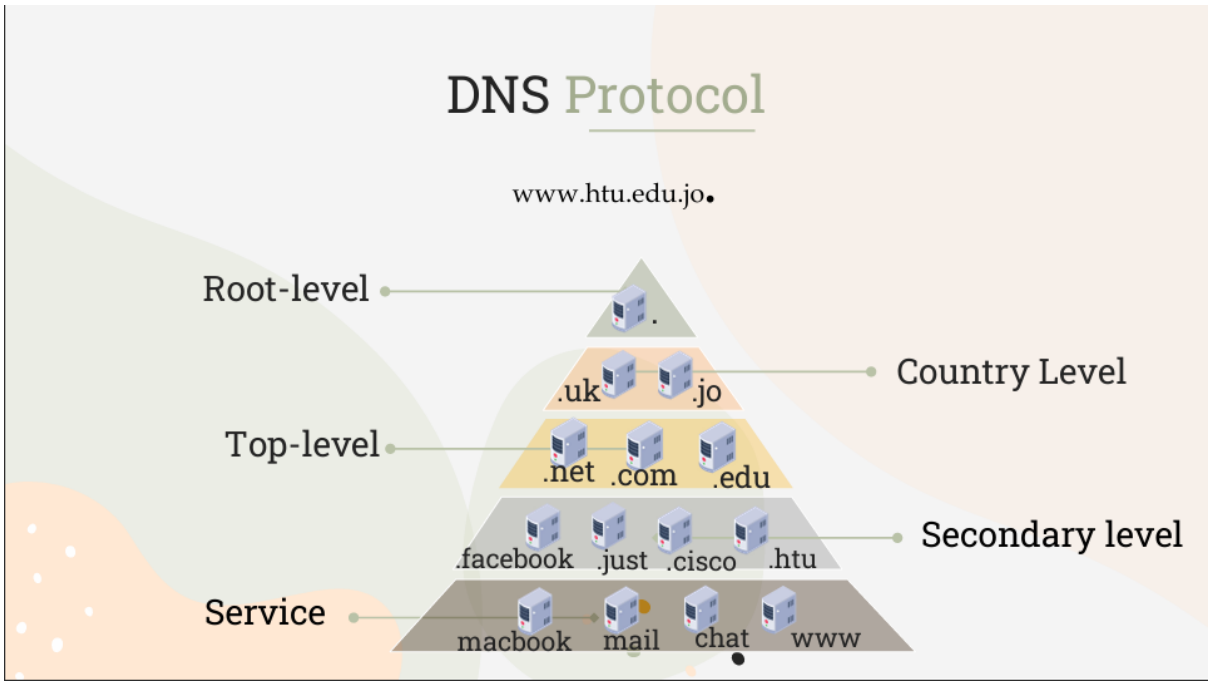
The root level domain is shown as an empty string or dot, and it is the highest level and starting point in the DNS hierarchy and managed 13 by root name servers worldwide.

The country-level domains are represented with the country code like jo, and it is responsible to indicate the location of the website, and they are managed by organizations designated by each country.

The top-level domains categorize domains based on their purposes like .com, or .org, it is the last words in the domain, and they are managed by many organizations such as ICANN.

Secon-level domains are the name of the website like google, and they are managed by the owner of the domain.

Service or host name is the first part of the domain name after the http like www, ftp, blog, etc., it shows the service or the type of the service, and it is configured by the domain owner.



2. Communication Protocols, Server Hardware, Operating System, And Web Server Software

2.1 Communication Protocols Definition

Communication protocols are a common set of rules that enable one or more device to communicate and transmit data over one or more networks. These protocols are standard in devices such as FTP, DHCP, and HTTP, etc. Each protocol has specific purpose and user that will be discussed below:

- TCP/IP (Transmission Control Protocol/Internet protocol): a protocol used to transmit data over a network, it prepares and forward these packets over the network.
- IP: The internet protocol is a protocol that do addressing for packets to route them to the destination.
- FTP (File Transfer Protocol): A protocol that is used for transferring files between client and server, for example, uploading website files to a web server.
- SMTP (Simple Mail Transfer Protocol): A protocol that is responsible for sending emails from a client to a server.
- POP (Post Office Protocol): A protocol that is responsible for retrieving these emails from a mail server.
- UDP (User Datagram Protocol): A protocol that serves in the transport layer and it provides fast, connectionless data transmission, and it is used for calls, video streaming, and online games.
- DHCP (Dynamic Host Configuration Protocol): A protocol that assigns Ip addresses automatically and dynamically for devices based on pools.
- HTTP (Hyper Text Transfer Protocol): A protocol that is responsible for accessing web applications.

2.2 Relationship Between Protocols, Hardware, Operating System, Web Server Software

2.2.1 Purpose And Relationship

Protocols enable one or more device to communicate and transmit data over one or more networks such as FTP, TCP/IP, UDP, etc.

Hardware are the physical components of a server like CPU, ram, NIC, etc., they provide the storage, and power to handle the requests.

Operating systems like Windows and Linux are the middleman between the hardware and software, it manages the hardware to make it able to run the software.

Web server software like Apache is the application that handles multiple requests like HTTP/HTTPS with a get, post, or put requests and the web server responds with a failure like (404) or success like (200).

After discussing the purpose of these components, we must discuss the relationship between them, the operating system is important to manage the hardware to make it able to run the software, without it, the hardware would not be able to run the web server software.

Web servers use the protocols like HTTP/HTTPS to communicate with the users and this web server runs on the operating system that provide the environment for the software to open.

The hardware are the main physical components to support the operating system, web server software, and protocols.

2.2.2 Relation To Designing, Accessing, and Publishing the Website

- Designing:
  For the protocols they are related to designing the website in multiple ways like if the website uses HTTP or HTTPS and to test the website, Also, I used FTP to transfer the website files from my computer to the hosting website.
  For the hardware I designed and tested the website on my own laptop, the hardware must be good to support the development tools.
  For the operating system, it provides the environment for these development tools, for me I used windows to develop the website on visual studio code.
  For the web server software, it makes a simulation of how the website will function on a live server, I used XAMPP for testing during the development.
- Accessing:
  The protocols are used in the accessing process, the users request the website, and their browser uses HTTP/HTTPS, Also, DNS is used in the accessing process to translate the domain name to Ip address.
  The hardware should be good to see the website content, hardware must be used to access the website to handle requests.
  The operating system must be reliable to make sure that the website still accessible, and it provides the environment to open these websites like the browser.
  The web server manages multiple requests like get, post, or put from the users, and it delivers the web page to the user's browser.
- Publishing:
  For publishing, FTP is used to upload the website files on the hosting site, and DNS makes sure that the domain name refers to the correct Ip address.
  The hardware is essential to upload these files and host the website on a physical or virtual server to handle the traffic.

The operating system makes sure that the web server software can manage the resources.
And the web server software shows the requested page to the user's browser.

3. Website Development Technologies, Tools, Techniques, And Frameworks

3.1 Website Development Technologies

The technologies are any invention, findings, innovations to solve problems, achieve goals, and create product. And it includes any development that have copyright protection or other protection under the law.

3.1.1 Available Website Technologies

- Browsers: browsers like edge, chrome, or brave allows users to access the website, and these websites have a difference in the way of rendering the website so older versions of browsers and some versions may not support some tags or styles.
- HTML and CSS: HTML manages the content, while CSS manages the styles. CSS animations improve user engagement, while placing HTML content correct makes the website easy to use.
- Programming Languages: programming languages like JavaScript allows us to add a dynamic validation on the sign in and sign up.
- Web development frameworks and libraries: frameworks like bootstrap provide reusable components and reduce time for developers.
- Databases: databases make sure that all data are placed correctly within multiple users, and all the data we need are placed in the database.
- Protocols: protocols like HTTPS makes sure that the communication is secure, and the sensitive data are protected
- Data Formats: data formats make it easier to know what the file is, and it improves management.
- APIs: it makes it easier to add a new feature or service to the website.

3.1.2 Technologies Used to Design, Develop and Manage Your Website

In my website I used various technologies to design, develop, and manage such as HTML and CSS, JavaScript, PHP, browsers, frameworks, and database.

I used HTML and CSS for displaying the content in my website and styling the content including colors, and fonts. I used various HTML to make users able to input data and I wrote the code with the best practices to make it easier for me to maintain and update, and I used CSS hover effects for buttons, and links which improves user engagement. Also, I used JavaScript for showing alerts when the user fills invalid username or passwords or when the user press checkout, and PHP was used for back-end development, I used PHP to make a connection with the database and to insert, read, update, delete from and to the database ensuring the website is dynamic. Database was used to store these data and manages Takke's data. Another technology I used to design and develop my website is bootstrap, I used bootstrap cards in my website, it saved time for me, and it made the code easier to maintain. The browser I used was Firefox Developer Edition to test the website design, functionality, and management.

3.2 Website Development Tools

Tools are the software's that are used for developing and designing the websites and they make it easier for developers to maintain their developing environment.

3.2.1 Available Tools for Website Design and Development

- Code Editors: these are a tools like visual studio code, and atom which they are used for writing and editing code, they have many extensions like prettier, live servers that allows developers to see the website live while writing code and it makes the code cleaner with colors. Also, it has debugging tools that helps the developers to identify errors.
- Browser Developer Tools: Firefox developer edition have many that allow testing screen sizes, network analysis tools, and console logs that show errors.
- Image Optimization Tools: these are the tools like Adobe illustrator that reduce file size and enhance quality which speeds up the website load.
- Wireframe/Sitemap Tools: these tools like Miro shows the whole design and functionality of the website, saving time and resources.
- Local environment: tools like Apache, PhpMyAdmin, and XAMMP allows the developer to test the website functions locally before publishing to the internet.
- WordPress: a tool that allows people to create a website without even have a code knowledge.
- Google Search Console: this tool sees the issues of broken links and improve search visibility, and it provide performance reports.

3.2.2 Tools Used to Design and Develop Your Website

There are many tools that were used in my website such as code editors, I used visual studio code and used the live server, and prettier extensions, and the debugging tools. Also, I used browser developer tools in Firefox developer edition browser, I used all the features I discussed above to make sure that everything works correctly in my website. Another tool used is Miro, I used Miro to design the wireframe and sitemap to provide the foundation for the website, it made me know the website hierarchy, and the wireframe made me put the design in my mind to a visual one, allowing me to be sure of the way of developing my website. The last tool I used which was the most crucial tool is the local development environment including XAMMP, Apache, PhpMyAdmin, these tools helped me to test all the functionalities of my website including reading from the database and insert, update, delete changes on the database, it helped me managing all database operations, while my computer was acting as a client and a server at the same time.

3.3 Website Development Techniques

It is a way or doing a specific task or a way of writing code such as responsiveness it means having a skill or ability in a field.

3.3.1 Available Techniques for Website Design and Development

- Responsive Design: It is a way that makes all website components work well on all devices, and it makes the developer don't make multiple versions for the website.
- Media Query: A technique that make sure all websites components and functionality adapt differently based on the device, and makes the developer don't maintain multiple codebase for different devices.
- Consistency Across Pages: A technique that improves user experience and make the website easier to navigate between pages.
- Accessibility: A technique that make sure that the anyone can use the website.
- Testing Early, Test Often: A technique that makes sure that the website functionalities work well reducing effort and cost.
- Title Tag: A tag that enhance SEO and the website appearance in browser tab.
- SEO indexing: A technique that make the website more discoverable and make sure that the website is visible to users
- SEO Meta Tags: A technique that improves the search engine rank and make it easier for browsers to index the website content.

- SEO Image Names and Alt Attribute: A technique that helps browsers to know what the content of images is and all images accessible to users.

3.3.2 Techniques Used in Your Website

In my website I used multiple techniques such as responsive design and media queries, which helped me making one version of code that work well on all device's screen sizes. Also, I used Consistency Across Pages, all page's links were implemented correctly and helps the user to an easy navigation between pages. Another technique used in my website is Testing Early, Test Often, it made sure that the website work as I want and fixed issues early, and it made sure that all functionalities work well, it reduced the effort. Also, I used the title tag in all pages that helps users to know in which page they are for example the home page title is Takke, and the cart page title is My Cart. Also, I used SEO Meta Tags like the description to improve the search engine website rank.

3.4 Website Development Frameworks

A framework is a technology that offers pre-built libraries, and they offer a way to create web applications, reusable components, and built-in functionalities.

3.4.1 Available Frameworks for Website Design and Development

- Bootstrap: A front-end framework that provide responsive components, reduces development time, provide reusable components, and make sure components look professional across devices.
- React.js: A front-end library that is responsible for building dynamic user interfaces, it improves reusability of components and simplifies maintenance.
- Angular.js: A front-end framework that is responsible for building dynamic webpages, it provides maintainability and simplifies the development of large websites.
- ASP.NET: A back-end framework that is responsible for API's and building web applications, it has a back end with high performance, and it simplifies maintenance.
- Django: A back-end framework that is responsible for building protected and secure web applications, it simplifies back-end development and provide admin panel.
- Laravel: A back-end framework that is responsible for building web applications with a great syntax, it simplifies back-end development and improve productivity with libraries and built-in components.

3.4.2 Frameworks Used in Your Website.

For the framework I used in my website, one framework was used which is bootstrap, I used the card component from bootstrap, it was used to reduce time and create a responsive design for this part. It was used to display the courses that users will see to add them to cart to display the content in a clean and clear way. Using the bootstrap card component made it easy for me to create a card design that is acceptable on all devices without using media queries for the card. Also, it reduced the development time, and it made it easier for me to maintain and update the code.

4. Search Engine Optimization

4.1 Search Engine Optimization Definition and Purpose

Search Engine Optimization is the process used to optimize website to make the webpage have a higher rank in the search engines so the pages can become easily findable and popular with user search, it improves the appearance and search engines rank them better.

The search engine optimization helps a website to rank higher in search results leading to more visitors, it configures the website design, navigation, and content making the website more user-friendly. Also, attracts visitors and target audience to visit the website, it helps the website to stay competitive and ranks higher than competitors.

4.2 Indexing Definition

Indexing is a concept in the Search Engine Optimization field, it is a process of collecting, analysing, and store information from website to include in the database, and this process is done by search engines.

The webpages that are indexed only appear in search engines but the webpages that are not indexed, they won't appear in search engines. Also, it helps the search engines to know and understand what is in the webpage and the content of the webpage is updated.

Indexing has many advantages such as make the pages visible in the search engines, the indexed pages will have a high rank in the search engine, and it makes sure that updates are reflected in the search engine.

4.3 Techniques For Improving the Ranking

There are a lot of techniques that can improve your website's ranking which they are very crucial for publishing the website, these techniques include:

1. Set Up Google Search Engine: google search engine a tool that helps the developers to monitor, maintain and troubleshoot of the website presence in google search engine. It has steps such as create account, add the URL of the website, submit a sitemap, and monitor the performance.
2. Create XML Sitemap: XML sitemap is a file that shows all the pages in the website, which helps the search engine to index webpages. The steps are generating sitemap using tools and

then submit the sitemap in google search console and in the code files.

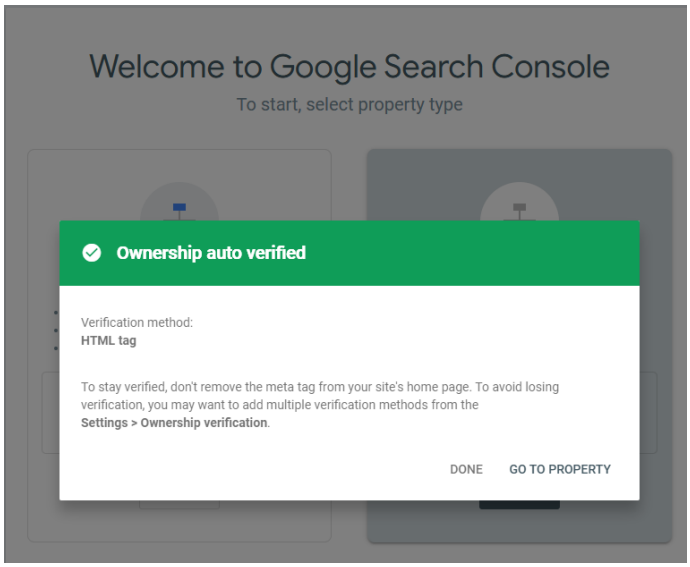
3. Include the title tag: the title tag is very crucial; it shows the title of the webpage that is shown in the browser tab and search engines.
4. Meta description tag: the meta description is a summarization of what the page contains, it is shown in the search engines below the title, the meta description should be unique and attract user to open the webpage.
5. Use of headings: headings are necessary in the webpage, and it improves readability, only one H1 tag should be in all pages, and the other levels of headings are used for the sections and subsections.
6. The use of alt attribute: the alt attribute shows a description of for an image, it helps screen readers to describe the image and it enhance user experience.
7. Fix broken Links: All links in the website should be working correctly, because the broken links lead to non-existent pages and it lower the website rank.

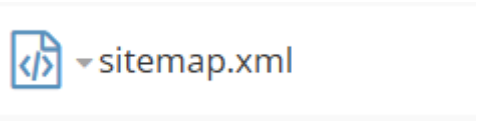
By applying these techniques in your website, your website will rank higher on the search engine, and leads to a better visibility.

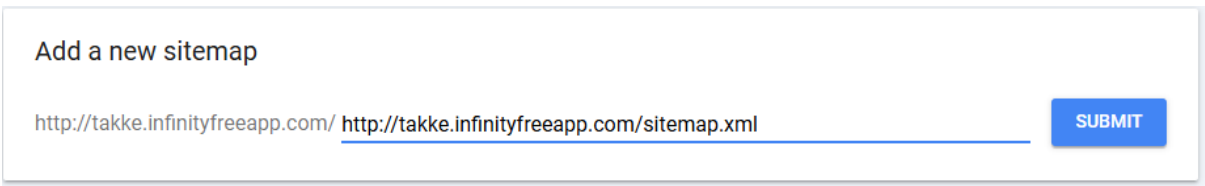
4.4 Evidence For the Used SEO Techniques in Your Website

I used many SEO techniques in my website including:

1. Set Up Google Search Engine and Create XML Sitemap: I set up google search engine and submitted a sitemap to it.







2. Include the title tag: I used the title tag in every page of my website, I choose a unique title for each page that describes the page in one word.

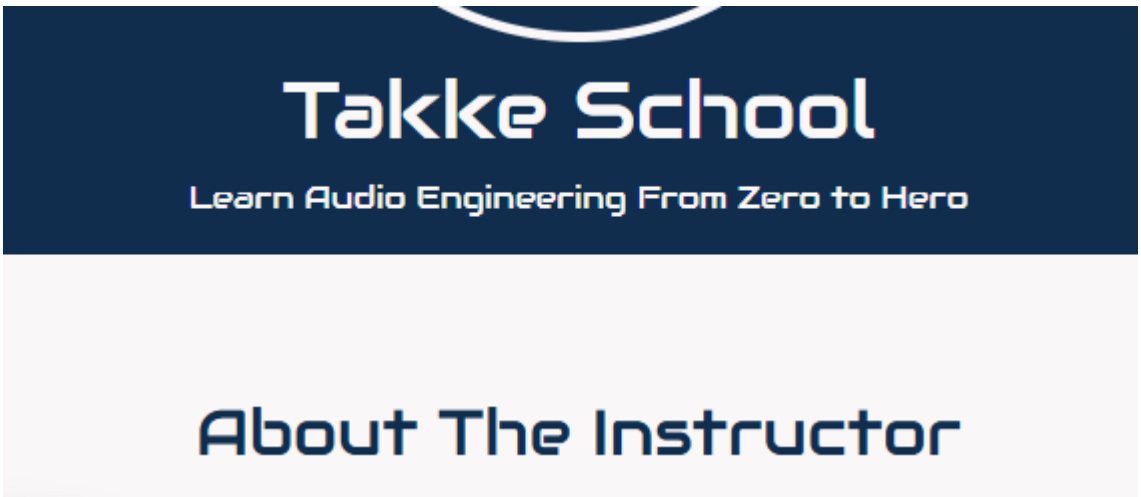
```
<title>Takke</title>
```

3. Meta description tag: I used the meta description tag in every page of my website, I choose a unique meta description tag for each page that describes the page in a small paragraph.

```
<meta name="description" content="Access your premium audio engineering courses on Takke. Dive into expert-led lessons on mixing, mastering,
and sound design. Continue your professional journey with our exclusive content.">
```

4. Use of headings: I used a lot of headings in each page with ensuring that there is only one h1 heading in each page for example "Takke School" is an h1 heading, while "about the instructor" is a h2 heading.

```
<div class="H1H">
  <h1>Takke School</h1>
</div>
<div class="Hparagraph">
  <p>Learn Audio Engineering From
</div>
</header>
<main>
  <div id="about">
    <div class="Htwo">
      <h2>About The Instructor</h2>
    </div>
```



5. The use of alt attribute: I used the alt attribute effectively for each image I added on the website.

```
<div class="headerImg">
  <img src="../Assets/Takkeh.png" alt="Takke logo - Premium Audio Engineering Courses">
</div>
<div class="H1H">
  <h1>Takke School</h1>
</div>
<div class="Hparagraph">
  <p>Learn Audio Engineering From Zero to Hero</p>
</div>
</header>
<main>
  <div id="about">
    <div class="Htwo">
      <h2>About The Instructor</h2>
    </div>
    <div class="AboutDetails">
      <img class="imgAbout" src="../Assets/instructor.png" alt="Takke audio engineering course instructor">
```

6. Fix broken Links: all links in my website work correctly and I tested each link.

```
<?php
if (isset($_SESSION['userName'])) {
  $user = $_SESSION['userName'];
  $checkQuery = "SELECT * FROM cart WHERE userName = '$user'";
  $checkResult = mysqli_query($con, $checkQuery);
  if (mysqli_num_rows($checkResult) > 0) {
    echo "
<a href='#courses'> <i class='fa-solid fa-store'> </i> Courses</a>
<a href='#about'> <i class='fa-solid fa-address-card'></i> About</a>
<a href='mycourses.php'> <i class='fa-solid fa-chalkboard-user'></i> My Courses</a>
<a href='cart.php'><i class='fa-solid fa-cart-shopping'></i>  My Cart </a>
<a class='headerAicon' href='editUserInfo.php'> <i class='fa-solid fa-user'></i> My Profile</a>
<a href='logout.php'><i class='fa-solid fa-power-off'></i> Logout</a>";
  } else {
    echo "
<a href='#courses'> <i class='fa-solid fa-store'> </i> Courses</a>
<a href='#about'> <i class='fa-solid fa-address-card'></i> About</a>
<a href='cart.php'><i class='fa-solid fa-cart-shopping'></i>  My Cart </a>
<a class='headerAicon' href='editUserInfo.php'> <i class='fa-solid fa-user'></i> My Profile</a>
<a href='logout.php'><i class='fa-solid fa-power-off'></i> Logout</a>";
  }
} else {
  echo "
<a href='#courses'> <i class='fa-solid fa-store'> </i> Courses</a>
<a href='#about'> <i class='fa-solid fa-address-card'></i> About</a>
<a href='signin.php'> <i class='fa-solid fa-chalkboard-user'></i> My Courses</a>
<a href='signIn.php'><i class='fa-solid fa-right-to-bracket'></i> Sign In </a>
<a href='signUp.php'><i class='fa-solid fa-right-to-bracket'></i> Sign Up </a>";
}
?>
```

5. Front-End And Back-End Website Technologies

5.1 Front-End And Back-End Definition and Related Technologies.

Front-End is the user side of the website, it is all the content that is visible to the user, it is the part where the user sees, and interact directly, it includes everything visible such as content, elements, styles, color palette, etc.

There are many technologies that are related to the front-end such as HTML, CSS, JavaScript, and frameworks like bootstrap.

Back-end is the server side of the website, it is everything that is nor visible to the user, it is the part that handles data including inserting data, updating data, reading data, and deleting data.

There are many technologies that are related to the back end such as PHP, databases, frameworks like Laravel, and API's.

Front end and back end are crucial in website development, they have a relationship that will be discussed below in the relationship section.

5.2 Presentation And Application Layers Definition.

The presentation layer is responsible for the user interface and communication layer, it has multiple functionalities such as presenting the data at the user's device. The presentation layer is where the user interacts with the application like it shows the graphical user interface, and common technologies are HTML, CSS, and JavaScript.

The application layer or the logic tier or middle tier provides the interface between the applications and the underlying network, it is the heart of an application, the presentation layer information that are collected are processed, and this application layer have the privilege to add, modify, or delete data in the data layer, and common technologies are python, java, and PHP.

5.3 Relationship.

In this part we will discuss of how the front-end and back-end relate to the presentation layer and application layer.

The front-end and back-end has a relation together, the front-end and back-end work together to create a full functional website with the visual data and the hidden data.

The front-end manages user-interface, interaction, and what the user sees, while the back-end manages these data, for example if the user wants to create an account he will se a sign up page, and when the user press on the sign up button, a hidden process happen in the back-end, it inserts the data that the user enters in the database.

There is a relation between front-end and the presentation layer, that is the front-end is the part of what the users see, and the presentation layer is responsible to display the interface for the user, the presentation layer formats and show the data to the user.

There ais a relation between back end and the application layer, the back-end handles the data, and the application layer collect the presentation layer information that were filled in the front-end part and process these data.

For example: a sign-up form was written in the front-end part, and the presentation layer formats the form and display it, the user fills the information and press sign up button, the application layer collect these information and the back end store these data in the database.

6. Online Website Creation Tools VS Custom-Built Websites

6.1 Online Website Creation Tools

There are many website creation tools such as WordPress, and wix, online website creation tools are a content management system that is used to create a fully operational website without the need of coding knowledge, you just drag components and drop them where you want in the website, and there are many themes and templets that people can use.

- WordPress: a website that is based on PHP and MySQL that anyone can use to build a website without coding required, but overloading a site with heavy themes may lead to slower the performance.
- Wix: a cloud-based website that anyone can create websites using drag and drop components and it's known for beginners, and it doesn't require coding.

The purpose of using online website creation tools is that it is easy to use for people who doesn't have a coding knowledge, it saves time because you take a little time to create the website and publish it, it is cost effective it doesn't take so much cost to build and it is affordable, it has many built-in tools like SEO features, some of the websites offers hosting services and domain management.

As there are many advantages and purposes for online website creation tools, there are also limitations such as the developer doesn't have a full control on the design and the layout, it doesn't have a unique designs and it only depends on templates that a lot of websites uses, scalability limitations, security concerns, and restricted functionality, as a developer you don't have all the access to the back-end functionalities or database.

6.2 Custom-Built Websites

Unlike website creation tools, custom-built websites are the websites that is developed from scratch, it requires a coding knowledge, and you can control every aspect of the website. In my opinion These websites are better than online website creation tool because the developer has a full control on the design and you can create complex layout and unique ones. Also, the performance here is optimized, you can write a clean code effectively to make sure the load time is fast and smooth. Also, you have all control on the functionality of the website, you can build custom features that are impossible with online website creation tools.

There are many tools that are used for custom-built websites including code editors and Ide's, these editors are used to write and edit HTML, CSS, JavaScript, and PHP code, for example there is Visual Studio Code which is the most famous text editor because it has a lot of features and provide extensions, also there is Atom text editor. Another tool is XAMPP and PhpMyAdmin which they are used to make the computer act as a server and a

computer at the same time, it makes a local environment for managing databases and back-end before publishing the website Also, the developer tools in browser which helps the developer to debug and test the website.

For the techniques, there are many techniques used for custom-built website such as responsive design, media queries, and test early, test often which I discussed them earlier in the report.

Also, there are many technologies such as HTML, CSS, JavaScript, databases, and protocols which I also discussed earlier in the report.

When I was developing the website, I made decisions on selection of technologies, management services, tools, and software to achieve the primary goal.

The technologies chosen were HTML and CSS, HTML was very important to choose to create the structure and content for my website, CSS was very important to style the content I added in the website, and to make a good visual content that user sees, JavaScript was chosen to show alerts and simulate the real-time validation. Also, PHP was chosen because it is very important in the back-end development, it made the connection between the database and the website, while database was chosen to store and manage website data, it was crucial to make the website manages users and other information's. Also, I choose bootstrap, and the cards displayed in the home page was bootstrap card because it saved some time and provided me with a pre-built component.

For the management services I first used XAMPP for local development, I used XAMPP to test the website before the publishing, XAMPP includes Apache, MySQL, and PHP which provided me with all the tools I needed for testing the website locally, it made my laptop work as a server and client at the same time. After testing the website on XAMPP, I choose infinityFree to host and publish my websites, I chose infinityFree because it provides a free hosting and the features for publishing the website.

Also, I used many tools like visual studio code, it was chosen because it offers an efficient development environment, it has features and extensions that helped me while writing and testing the code. Also, browser tools in Firefox developer edition were chosen due to its advantages, I tested the responsiveness and debugging and made sure that the website work well on all devices.

So, the software's used was:

- Visual Studio Code
- Firefox Developer Edition
- Miro
- XAMPP

6.3 Comparison With Regards to Design Flexibility, Performance, Functionality, User Experience (UX) And User Interface (UI).

There are a lot of differences between online website creation tools and custom-built sites including the design flexibility, performance, user experience, user interface.

Online website creation tools provides themes and templates for webpage design but for the custom-built sites, you can build the website with any design you want, you start from scratch, so unique and complex design can't be done with online website creation tools, as you have the full control on the design while developing a custom-built sites, so if the website has special design requirements you can't do these requirements with the online website creation tools.

For the performance, online website creation tools may have performance issue, these websites may come with overload and the developer may add so many components that impact the performance, but the custom-built site offers more optimization and faster performance as the developer write a clean and efficient code and the developer have more control over code optimization.

For the functionality, online website creation tools has plugins and libraries that offers fixed functionalities as the developer doesn't have full control over the functionalities and for complex functionalities, online website creation tools may be not possible to implement these complex functionalities, but for the custom-built websites, developers have a full control on the functionalities as they write these functionalities from scratch, and they can write a clean complex functionalities to meet a specific functionality requirements.

For the user experience, online website creation tools provide a good UX, but it provides poor customization and may not lead to the specific needs which provide poor UX, but for custom-built sites allows customization in the UX and make sure that it meets the needs, and you will have full control over the UX.

For the user interface, online website creation tools don't offer a unique interface, it provides an interface that is used by a lot of websites, unlike the custom-built websites that provide a unique interface that there is not any website that uses which allows full control on the UI.

In conclusion, we know that online website creation tools don't allow the developer to have a full control on the website, but the custom-built websites allows the developer to have a full control on the website.

Overall, this is a simple comparison:

	Online Web Creation	**Custom-built sites**
Design Flexibility	Limited	High
Performance	May be slower	Faster
Functionality	Limited to plugins	Custom functionalities
UX/UI	Poor	High

7. Design And Development Process

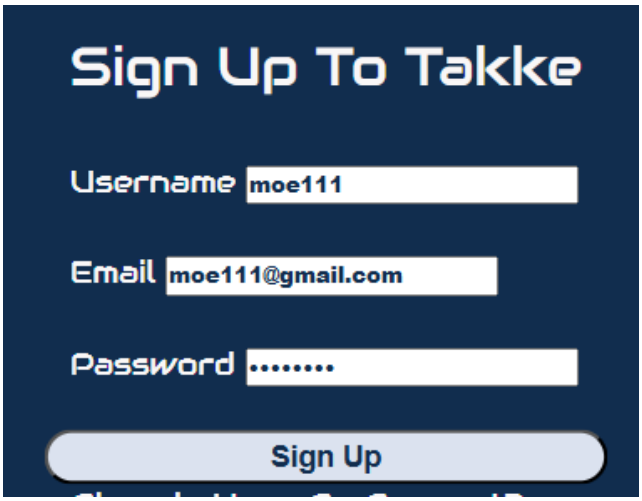
7.1 Website Vs. Design Document

The design document is basically a description of the design of the web application, the scope, the requirements, and what the final website will be.

For the goals that was set in the design document, my final website aligns well with the website goal as it offers paid sound engineering courses, and the primary goal was giving the students a comprehensive knowledge and high skills that not is popular because and there aren't many websites that offers sound engineering courses which the website shows that I achieved that goal by providing courses, guidance through email, ease of access, and clear and clean layout.

The design document shows the requirements, scope, goal, and the decisions I made during the development of Takke that were based on the design document requirements such as:

1. User Account Management: The design document shows a deep explanation of the user account management requirements, so to apply this requirement, I implemented a system that handles the sign up process that the users will be able to use to create accounts and store their information in the database, and for the sign in feature, it allows users to sign in and access their profiles, and sign out was implemented to clear the session data. As we can see if we sign up in the website, it insert the data in the database

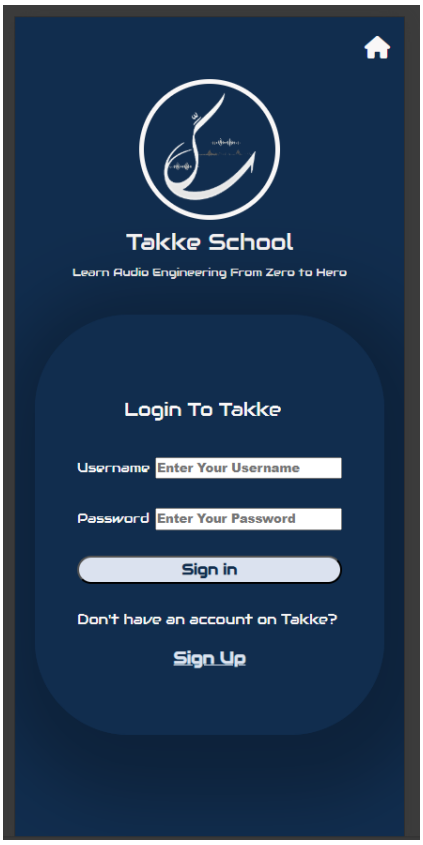


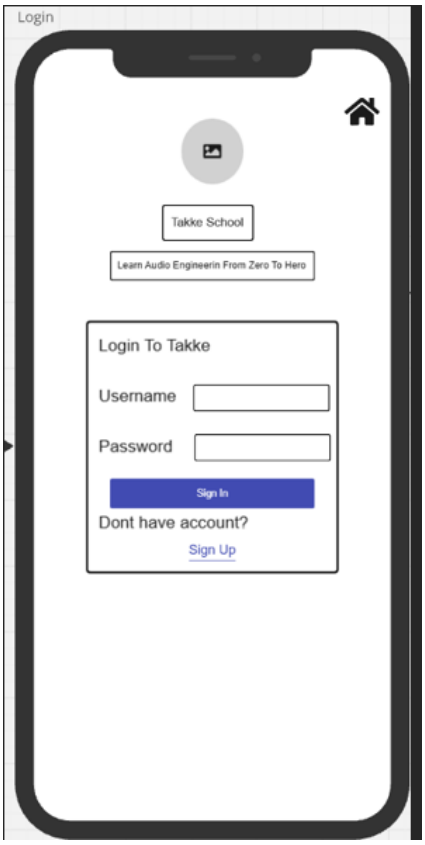


2. Dynamic Webpages: The design document shows that there is a need for 5-6 dynamic webpages which have the CRUD operations, in the final website, users can insert data to the database by creating an account, the courses are read from the database and displayed in the home page, users can update their data from the profile web page, and users can delete a course from the cart and removed from the cart table in the database and we have which all matches the design

document. I have 5 webpages in my website which are Takke home page, my cart page, my courses page, sign in page, and sign-up page.

3. Responsiveness Design: the design document shows that the website should be accessible well on all devices such as phones, laptops, tablets, and pc's. So, I used CSS flex layout, and I implemented media query to make the website matches exactly the wireframe and work well on all devices.
4. Wireframe and SiteMap: By making all of these decisions and applying all of the requirements I was successfully able to create a website that matches exactly the wireframe for mobile devices and for desktops, and the website have the same pages as the sitemap.





In conclusion, the implementation of the website aligns well with the design documents by achieving the goals, scope, requirements, sitemap, and wireframe that matches the website. By implementing all of these decisions I was successfully able to meet user expectations and my expectations effectively.

7.2 Evaluation and Analysis

All of the design and development decisions of Takke were based on the design documents and the requirements to make sure that the final website achieve the goals and scope and matches everything in the design document.

Before I started developing, I started with analyzing goals and scope.

The website goal was creating a unique learning website for aspiring sound engineers. The implementation of the website meets the goal, offers paid sound engineering courses that cover the sound engineering aspects, provide guidance through email, and making sure that the user interface is clear and clean, and the website showed a deep knowledge of the sound engineering field by making it one of the few resources that offers sound engineering courses.

Takke successfully implemented the initial scope that was discussed in the design document, it delivers an educational resource through clean layout, it ensures that the website work well on different device's screen size. By meeting the scope of Takke, Takke is a user-friendly website that attracts users to see and buy.

The final website matches the wireframe and sitemap which I showed earlier.

The website focuses on applying all the requirements, by making the database integration essential all back-end functionalities were implemented well and effectively, and all five pages were connected to the database and functioned dynamically which reflected the design document.

After discussing the design and development process of the multipage website against the design document, I must discuss the issues and challenges faced, and any area of improvements.

I had many challenges while designing and developing the website, the designs and responsiveness were not implemented correctly as the wireframe at first which took me a lot of time to debug, the solution was that when I included the CSS link, I entered a different link that made the designs not showing in my website.

Also, I had a challenge where the inserting courses to the cart table wasn't working which was because incorrect SQL query which I was using I different table name.

There was the crucial challenge for me which was translating the wireframe into a fully functional responsive website and maintaining the back-end process.

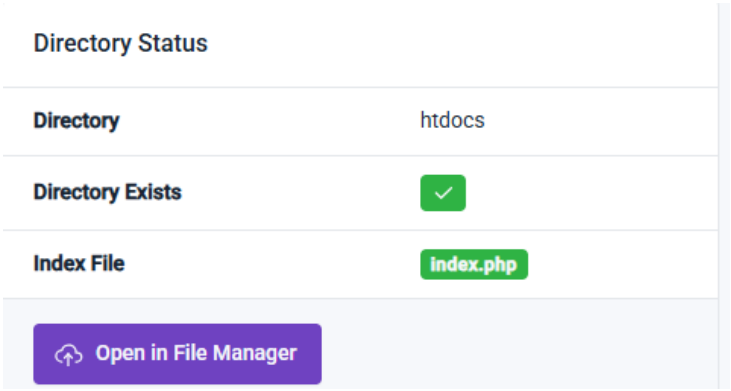
For me, I have many areas for improvement to make the website better such as improving the security of the website by making an SSL certificate and do hashing, and encryption of data. Also, I may add many different tools that helps the student to monitor their performance such as adding progress bar, assignments, more courses.

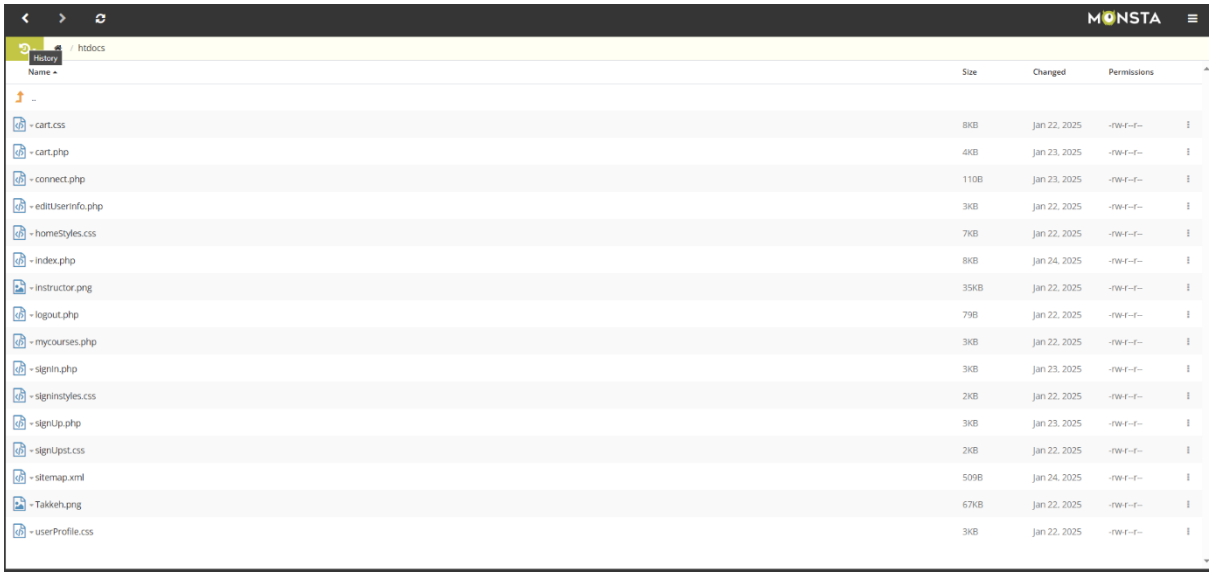
In conclusion the design and development of Takke met the design document's goals, scope, requirements, sitemap, and wireframe successfully. The challenges and issues I face were solved effectively through debugging and testing.

Hosting (http://takke.infinityfreeapp.com/)

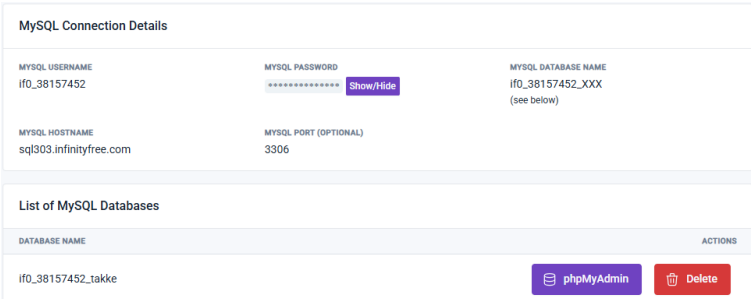
For Hosting, I used InfinityFree because it offers free hosting plans, and it supports for PHP and MySQL databases.

I configured the domain Takke as the website brand's name, and I used File Manager, using FTP to upload my website. And I configured the database in the infinityFree by creating a new database and then I imported my original database.









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